

The Complement Rule Is Excluded Middle: Machine-Checked Probability over Intuitionistic Logic

Anonymous Author(s)

Abstract

Kolmogorov’s probability is defined within a classical logic framework, whose complement rule becomes equivalent to the law of excluded middle in the limiting case of certainty. That framework presupposes every event is decidable, which fails for the semidecidable properties, termination, reachability, that verification cares about. We mechanize, in Lean 4 over mathlib, what probability becomes when the event logic is *intuitionistic*: valuations on frames, the algebras of verifiable propositions. Two characterizations anchor the development: (1) the complement rule $v(\neg A) = 1 - v(A)$ ¹ holds for every valuation exactly when the algebra satisfies excluded middle, and (2) Dempster conditioning agrees with Bayesian conditioning at a piece of evidence exactly when that evidence carries zero slack. On finite frames, every valuation is a classical mass function on points. In the general case, a valuation splits into atomic and diffuse parts, and its Boolean core carries a totally monotone Dempster–Shafer belief function. We define the slack of an event A as $1 - v(A) - v(\neg A)$: in the measure model it is the boundary mass. On a semidecidable event, forcing it to zero is exactly as hard as solving the halting problem. We further prove that product valuations are unique on chain products and provably non-unique on the powerset frame, where Fubini fails, show that the naive transcription of Cox’s theorem is unsatisfiable, and build the Aczél/Hölder analytic core from scratch.

Keywords: intuitionistic logic, probability, Cox’s theorem, belief functions, locales, Lean, mathlib, formalized mathematics

1 Introduction

Kolmogorov’s probability [Kolmogorov 1933] assigns numerical values representing uncertainty to events. The events to which the probability is assigned form a Boolean σ -algebra, therefore satisfying the law of excluded middle. The axioms these assignments satisfy make up the arithmetic of chance (non-negativity, normalization, σ -additivity).

This framework commits to logical axioms that computer science cannot always commit to. Many properties of programs (does this computation halt, does this process reach a given state) are only *semidecidable*: a procedure confirms the property when it holds, and no procedure, in general, confirms its failure. The natural event algebra for a semidecidable property is a *frame*, the algebra of open sets of a

space, and equivalently the algebraic semantics of intuitionistic logic. We show that applying Kolmogorov’s probability (with its underlying Boolean event algebra) to a semidecidable property presupposes a solution to the halting problem. This result is the pointwise counterpart of Chaitin’s Ω [Chaitin 1975] example, which we interpret in the framework we develop here. In general, reasoning about the chance that a program has some property therefore needs an event logic that is not Boolean. We develop a mechanized framework that allows us to study probability calculi without committing to a fixed (classical) logic, and thereby select the probability calculus aligned with the logic of the objects it reasons about.

Probability is, as the Cox–Jaynes tradition has it, *extended logic* [Cox 1946; Jaynes 2003]. Different logics can be extended through the same construction. First, fix an event algebra, i.e. the chosen event logic in algebraic form. This algebra consists of the underlying set of events, together with the operations interpreting the logic’s connectives. For the logics compared in this paper these are conjunction, disjunction, and negation, with the law of excluded middle holding in the classical case. Then let *valuations* be the maps from events to numbers that satisfy the axioms of monotonicity, normalization, and modularity. The resulting calculus is a probability calculus (i.e. degrees of certainty rather than an arbitrary grading of events) when (1) the valuations taking only the values 0 and 1, the two degrees of complete certainty, reconstruct the underlying event logic, and (2) the construction, evaluated at classical logic, returns the classical theory. This construction treats the logic of events as a *parameter*: a choice of event algebra determines an associated probability calculus. In this work, however, we do not treat the general case: there is no guarantee that *each* event algebra returns a calculus of degrees of certainty.

Kolmogorov takes the event algebra to be Boolean, and his calculus is the value of the construction at Boolean logic. There, the law of excluded middle in the underlying logic makes it possible to show that every valuation obeys the *complement rule* $P(A^c) = 1 - P(A)$. A motivating example for varying the logic parameter is quantum uncertainty calculus, whose event algebras are the orthomodular lattices [Birkhoff and von Neumann 1936; Gleason 1957]. Here, we study the construction extending intuitionistic logic, whose event algebras are frames: complete Heyting algebras, the lattices of open sets. (1) The all-or-nothing valuations are exactly the two-valued models of the logic (Section 3). (2) Complete

¹By $\neg a$ we mean the pseudocomplement of a in a frame.

Boolean algebras are exactly the frames satisfying excluded middle, so the classical theory embeds as a special case.

Given a valuation $v : \Omega \rightarrow [0, \infty]$ on a frame Ω , for all a, b , by modularity, $v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b)$. The pseudocomplement $\neg a$ satisfies $a \sqcap \neg a = \perp$ but not, in general, $a \sqcup \neg a = \top$, so the complement rule degrades to $v(a) + v(\neg a) \leq 1$, with a gap

$$\text{slack}(a) = 1 - v(a) - v(\neg a).$$

Classically, this is the width of the Dempster-Shafer belief/plausibility interval [Dempster 1967; Shafer 1976], quantifying ignorance. In our system it instead quantifies the mass of the part of a that is neither confirmed nor refuted, the probabilistic image of the semidecidable gap above. On the opens of a space with measure μ , the slack of an open U is $\mu(\partial U)$, the mass of its topological boundary. For verification this matters concretely: knowing how likely a property is to hold no longer tells you how likely it is to fail, and the gap between the two, the slack, is now also required to answer that question.

How the change of logic affects statistical reasoning can be stated operation by operation. The conditional $v(a \sqcap b)/v(b)$, the product rule, and Bayes symmetry hold verbatim. The classical denominator of Bayes' theorem, $P(B) = P(B \mid A)P(A) + P(B \mid \neg A)P(\neg A)$, degrades to a lower bound whose deficit is a conditional slack. Marginalization is exact only over partitions. The complement trick $P(A) = 1 - P(\neg A)$ degrades to the interval $[v(A), 1 - v(\neg A)]$ of width $\text{slack}(A)$. Dempster's rule and Bayesian conditioning, classically indistinguishable, give different posteriors from the same evidence unless its slack vanishes. The value of an event A itself is read as $P(\text{int } A)$, so a nonatomic distribution queried on intervals still behaves classically: an interval's boundary is two points, carrying no mass. The two readings differ only at boundaries carrying mass and at semidecidable hypotheses.

Cox's original theorem required repair after Halpern's counterexample [Cox 1946; Halpern 1999; Horn 2003]. Even with this repair, the direct translation of Cox's theorem to intuitionistic logic is *unsatisfiable*. This is because the sum rule is irreducible: a disjunction's value is provably not a function of its disjuncts' values, so modularity must be posited rather than derived. Weak excluded middle, an assumption strictly between intuitionistic and classical logic, does not restore the reducibility of the sum rule: the counterexample frame itself satisfies weak excluded middle.

Contributions.

1. **Weak excluded middle** (Sections 2 and 7). Global weak excluded middle on the opens of X is exactly extremal disconnectedness of X , for compact Hausdorff X exactly the Stonean spaces.
2. **Two characterizations** (Sections 3 and 5). The complement rule holds for *every* valuation iff the frame satisfies excluded middle, the exact content of Van Horn's

classical negation axiom R3 [Horn 2003]. Dempster and Bayesian conditioning agree iff the evidence has zero slack.

3. **Structure theorems** (Sections 4 and 5). On finite frames a valuation is a classical mass function on points. In general it splits into atomic and diffuse parts, and on its Boolean core it is a belief function.
4. **Cox's theorem, dissected** (Section 6, Appendix C). The product-rule half is independent of the logic, proved by an Aczél/Hölder generator built from scratch. The sum-rule half, a theorem classically, admits no derivation here (a five-element counterexample) and must be posited.
5. **Two grounding models** (Section 7). Any classical measure, read on the opens, is a valuation whose slack is boundary mass. A slack-free readout of halting is uncomputable, and on Cantor space the value of the halting event is Chaitin's Ω . Slack is submodular yet does not determine the valuation.
6. **A first formal treatment**. Everything above is machine-checked in Lean 4 over mathlib, to our knowledge the first mechanization of non-additive probability over intuitionistic logic.

Mechanization. The development is carried out in Lean 4 [de Moura and Ullrich 2021] over mathlib [mathlib Community 2020], in roughly 5,000 lines with no sorry, and is included in full as anonymous supplementary material.² Two points are specific to a proof assistant. We separate *object-level* classicality, excluded middle in the frame Ω , from the *meta-level* classicality mathlib inherits through `Classical.choice`, track which results use which, and equip the one essential non-Boolean model with a computable decidability instance. The product-rule half of Cox's theorem needs an Aczél/Hölder regratuation absent from mathlib, which we build from scratch.

2 Valuations on Frames

2.1 The Event Logic

Definition 1 (frame). A *frame* is a complete lattice Ω in which finite meets distribute over arbitrary joins,

$$a \sqcap \left(\bigsqcup_{i \in I} b_i \right) = \bigsqcup_{i \in I} (a \sqcap b_i)$$

for every $a \in \Omega$ and every family $\{b_i\}_{i \in I} \subseteq \Omega$. Equivalently, it is a complete Heyting algebra. By $x \Rightarrow y$ we denote the Heyting implication in a frame, and $\neg a := a \Rightarrow \perp$ is the pseudocomplement of $a \in \Omega$. Also, a^c denotes a complement of a , if it exists, i.e. an element such that $a \sqcap a^c = \perp$ and $a \sqcup a^c = \top$.

²Claude (Anthropic) was used to assist with drafting and debugging the Lean formalization.

Frames are the algebraic semantics of intuitionistic logic and the point-free abstraction of topological spaces. The same objects with morphisms reversed form the category of *locales*, so the distinction is invisible at the level of a single algebra. We say *frame* throughout and reserve *locale* for the spatial and categorical statements (points of a locale, classifying locales) where that word is the standard one. The motivating example throughout is $\Omega = \mathcal{O}(X)$, the opens of a space, with $\sqcap, \sqcup$ as intersection and union and the pseudocomplement $\neg a$ of an open given by the *interior* of its set complement. In mathlib this is the typeclass `Order.Frame`, and the Heyting operations come with the key asymmetry already in place: $a \sqcap \neg a = \perp$ holds in any Heyting algebra (`inf_compl_eq_bot`), while $a \sqcup \neg a = \top$ is available only in Boolean algebras. An element with $a \sqcup \neg a = \top$ is *complemented*: it satisfies its own instance of excluded middle.

2.2 The Valuation Structure

The central object assigns a degree of certainty to each element of the frame, with inclusion–exclusion as the only interaction axiom:

Definition 2 (valuation). A *valuation* on a frame Ω is a *monotone* function $v : \Omega \rightarrow [0, \infty]$ with $v(\perp) = 0$, $v(\top) = 1$ (*normalization*), and, for all $a, b \in \Omega$, *modularity*:

$$v(a) + v(b) = v(a \sqcup b) + v(a \sqcap b).$$

```
structure Valuation (Ω : Type*) [Order.
  Frame Ω] where
  toFun : Ω → ℝ≥0∞
  map_bot' : toFun ⊥ = 0
  map_top' : toFun ⊤ = 1
  mono' : Monotone toFun
  modular' : ∀ a b, toFun a + toFun b
    = toFun (a ⊔ b) + toFun (a ⊓ b)
```

This axiom set is not ours: Weatherson’s axioms for intuitionistic probability are exactly these [Weatherson 2003], and the valuations of constructive measure theory [Coquand and Spitters 2009; Vickers 2008] are their Scott-continuous instances (Section 8).

Values live in the extended non-negative reals $[0, \infty]$, and normalization confines them to $[0, 1]^{\text{Lean}}$. The conventions this choice carries, unconditional summation and truncated subtraction, are an artifact of the mechanization and are recorded in Appendix A.

2.3 Slack

Modularity applied to the disjoint pair $(a, \neg a)$ collapses to $v(a) + v(\neg a) = v(a \sqcup \neg a)^{\text{Lean}}$. So, the join need not be $\top$:

Theorem 3 (the complement bound). *For every valuation v on a frame Ω and every $a \in \Omega$, $v(a) + v(\neg a) \leq 1$.*

```
theorem Valuation.add_compl_le_one (v :
  Valuation Ω) (a : Ω) :
  v a + v a^c ≤ 1
```

This is a *derived* Dempster–Shafer sub-additivity inequality. The deficit

$$\text{slack}(a) = 1 - (v(a) + v(\neg a)) = 1 - v(a \sqcup \neg a)$$

is the valuation of the undecided region. It decomposes canonically. The pair $(\neg\neg a, \neg a)$, double negation against negation, is always disjoint. However, it is a partition only under *weak* excluded middle ($\neg a \sqcup \neg\neg a = \top$). Applying modularity to this disjoint pair splits the slack into two logically independent gaps:

Definition 4 (regular element). An element $a \in \Omega$ is *regular* (or $\neg\neg$ -stable) if $a = \neg\neg a$.

Theorem 5 (slack decomposition^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$, $\text{slack}(a) = \text{dnGap}(a) + \text{deMorganGap}(a)$, where $\text{dnGap}(a) = v(\neg\neg a) - v(a)$ is 0 if a is $\neg\neg$ -stable (regular), and $\text{deMorganGap}(a) = 1 - v(\neg\neg a \sqcup \neg a)$ is 0 if weak excluded middle holds at a .*

The single Dempster–Shafer “ignorance” number is thus resolved into a regularity defect and a De Morgan defect: the first measures failure of double-negation stability, the second corresponds to a named intermediate logic (weak excluded middle). Constructively, the contrapositive of a universal hypothesis returns only its double-negated form, so for open sets, this specification of probability does not fall victim to Hempel’s raven paradox [Hempel 1945]: evidence of the white sock indoors confirms at most $\neg\neg a$, not a itself, and dnGap is exactly the gap between confirming $\neg\neg a$ and confirming a itself.

Because dnGap and deMorganGap are themselves defined from v , the apparent three unknowns $v(a)$, $v(\neg a)$, and $v(\neg\neg a)$ are not independent: they are part of a three-constraint system, yielding the following result:

Theorem 6 (slack in closed form^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$,*

$$\text{slack}(a) = (v(\neg\neg a) - v(a)) + (1 - v(\neg\neg a) - v(\neg a)),$$

where $v(a) \leq v(\neg\neg a)$.

The inequality follows from monotonicity applied to $a \leq \neg\neg a$, a fact of every Heyting algebra, and shows the first summand is exactly the difference, not the floor produced by $[0, \infty]$ ’s truncated subtraction. The split into a regularity defect and a De Morgan defect follows from monotonicity and modularity alone, with no further axiom about ignorance required. This contrasts with the usual Dempster–Shafer picture, where the mass assigned to non-singleton, ambiguous sets is an extra primitive.

Theorem 6 expresses slack pointwise in terms of v . The converse, whether slack in turn determines v , is more delicate. One constraint is automatic:

Theorem 7 (slack is submodular^{Lean}). *For every valuation v on a frame Ω and every $a, b \in \Omega$,*

$$\text{slack}(a \sqcup b) + \text{slack}(a \sqcap b) \leq \text{slack}(a) + \text{slack}(b).$$

The proof runs v 's own modularity twice, once at (a, b) and once at $(\neg a, \neg b)$, and combines the two results with the De Morgan laws: the equality $\neg(a \sqcup b) = \neg a \sqcap \neg b$, which holds in every Heyting algebra, and the inequality $\neg a \sqcup \neg b \leq \neg(a \sqcap b)$, monotone under v but not in general an equality. Any function proposed as the slack of some valuation must therefore be submodular.

Remark 8 (non-recovery^{Lean}). Submodularity is necessary, not sufficient: Section 7.2 exhibits two valuations on the same frame with identical slack functions.

Remark 9 (slack under weak excluded middle^{Lean}). The two gaps, dnGap and deMorganGap , computed at a single element, are logically independent. However, if weak excluded middle is imposed on the frame for every a ($\neg\neg a \sqcup \neg a = \top$ ^{Lean}), then deMorganGap vanishes identically and $\text{slack}(a) = \text{dnGap}(a)$ for every valuation and every a . This sits strictly between plain intuitionistic logic, where both gaps can be positive, and full excluded middle, where both vanish^{Lean}. A concrete instance appears in Section 7.

2.4 What Survives without Excluded Middle

Three classical pillars survive intact, and one fails in a precisely delimited way. *Disjoint additivity* is unconditional: $a \sqcap b = \perp$ implies $v(a \sqcup b) = v(a) + v(b)$. *Conditioning* is native: for $v(b) \neq 0$ the posterior $v(a \mid b) = v(a \sqcap b)/v(b)$ is again a valuation (the proof is frame distributivity), and the product rule $v(a \mid b) \cdot v(b) = v(a \sqcap b)$ and Bayes symmetry (and therefore Bayesian updating) $v(a \mid b)v(b) = v(b \mid a)v(a)$ hold^{Lean}. *Total probability* holds over any genuine partition, meaning $a \sqcap a' = \perp$ and $a \sqcup a' = \top$: $v(b) = v(a \sqcap b) + v(a' \sqcap b)$. What fails is total probability over $\{a, \neg a\}$. That family need not join to $\top$, and the conditional masses fall short of $v(b)$ by a conditional slack^{Lean}.

The classical fragment. We characterize the elements of Ω on which the classical complement rule holds. The $\neg\neg$ -stable (regular) elements, which form a Boolean algebra (by Glivenko's theorem), are a natural candidate, but they do not suffice. Modularity constrains the frame join $\sqcup$, whereas the Boolean join of regular elements is the regularized $\neg\neg(a \sqcup b)$, for which only the inequality $v(a) + v(b) \leq v(\neg\neg(a \sqcup b)) + v(a \sqcap b)$ survives (i.e. the 2-monotonicity of Section 5). A regular but uncomplemented element, which is an open ray $(-\infty, 0)$ in the opens of $\mathbb{R}$, under an atomic measure at 0, still carries slack. The correct classical fragment is the *complemented* elements, where $v(a) + v(a^c) = 1$ holds exactly^{Lean}. On a connected locale that fragment can be as small as $\{\perp, \top\}$. In the Boolean limit, every element is complemented, the slack vanishes identically, and Kolmogorov is recovered^{Lean}.

Mixtures. Valuations are closed under finite convex combination^{Lean}, with every axiom inherited termwise, because modularity is linear. This innocuous fact becomes a characterization in Section 4.

3 Classicality in Theory and Metatheory

Classicality enters this work twice: in the metatheory, the logic of the proof assistant itself, and in the object theory, the frame's own logic.

Lean's meta-theory. Lean's foundations include `Classical.choice`, and `mathlib` is built on it throughout. For example, the order relation on $\mathbb{R}$ and $[0, \infty]$ has no computable Decidable instance in general, so any library lemma that case-splits on $x \leq y$ for reals resolves that split via `Classical.propDecidable`. A formalization that never invokes excluded middle at the level of frame elements $a \in \Omega$ can therefore still be classical through the arithmetic it borrows from the library. We track this separately from object-level classicality throughout the paper. Theorem 20 of Section 4 rests only on this ambient baseline and contains no explicit case split. The Scott-continuity recovery theorem (Theorem 19) invokes Lean's classical tactic directly, and the explicit Cox model of Section 6 goes the other way, replacing the default with an explicit computable Decidable instance to keep that one instance genuinely non-classical.

Object-level classicality: the R3 axiom. Van Horn's repaired Cox theorem [Horn 2003] axiomatizes plausibility over classical propositions. Its negation axiom, called *R3*, posits that the plausibility of $\neg A$ is a fixed non-increasing function S of the plausibility of A . Consistency forces the involution $S \circ S = \text{id}$, and therefore the complement rule $p \mapsto 1 - p$. We show that the involutive negation is double-negation elimination, so *R3* is where classical logic enters every known Cox-style derivation. To do this, we first isolate the exact content of *R3*'s conclusion as a predicate on valuations, then prove equivalence to excluded middle.

That predicate is *classical negation*: a valuation v has classical negation if $v(a) + v(\neg a) = 1$ for every $a \in \Omega$.

Theorem 10 (the *R3* characterization^{Lean}). *For every frame Ω , every valuation on Ω has classical negation iff $a \sqcup \neg a = \top$ for every $a \in \Omega$.*

The forward direction is a calculation. The content is the converse: wherever excluded middle fails, a valuation under genuine slack must be *constructed*. Suppose $a \sqcup \neg a \neq \top$.

Definition 11 (ideal, prime ideal). An ideal $J \subseteq \Omega$ is downward closed and closed under finite joins. J is *prime* if $\top \notin J$ and $a \sqcap b \in J$ implies $a \in J$ or $b \in J$.

The prime ideal separation theorem (`mathlib's DistribLattice.prime_ideal_of_disjoint_filter_ideal`)

yields a prime ideal J containing $a \sqcup \neg a$ but not $\top$. The indicator of J 's complement,

$$v(x) = \begin{cases} 0 & x \in J \\ 1 & x \notin J, \end{cases}$$

is a valuation. Modularity in the only nontrivial case ($x, y \notin J$) is *exactly* primeness, $x \sqcap y \in J \Rightarrow x \in J \vee y \in J$. And this valuation assigns $v(a) + v(\neg a) = 0 \neq 1$.

Sharp valuations are the points of the locale. The construction above is one half of an exact correspondence. Call v *sharp* if it is $\{0, 1\}$ -valued, a state of full certainty. Sharpness characterizes exactly the prime-ideal indicators:

Theorem 12 (sharp valuations are points^{Lean}). *For every valuation v on a frame Ω , v is sharp iff v is the complement-indicator of a prime ideal.*

The correspondence is given by two mutually inverse maps: $v \mapsto Z(v) = \{x \in \Omega : v(x) = 0\}$, which computes the zero-set of a sharp valuation, and the map taking J to its complement-indicator (the valuation of Theorem 10's construction). $Z(v)$ is prime exactly because v is sharp, and the complement-indicator of $Z(v)$ recovers v itself. In locale-theoretic terms, sharp valuations recover the *points* of the locale. A prime ideal is *completely prime* when it is also closed under arbitrary (not just finite) suprema of its members. The finitely prime points of Theorem 12 sharpen to the completely prime points exactly where Scott continuity is added, recovering points of the locale in the sense used by [Coquand and Spitters 2009; Vickers 2008]:

Theorem 13 (Scott-continuous sharp valuations^{Lean}). *For every valuation v on a frame Ω , v is sharp and Scott-continuous iff v is the complement-indicator of a completely prime ideal.*

The characterization and the two correspondences together locate the object-level classicality precisely: whenever excluded middle fails at some a , the failure occurs at a point of the locale.

4 Valuation Representation

Every valuation on a finite frame is a probability mass function on points. On infinite frames, this can fail, and Scott continuity marks the boundary between where it still holds and where it does not.

4.1 Finite Frames Representation

Theorem 14 (Birkhoff duality [Birkhoff 1937]). *Every finite distributive lattice is isomorphic to the lattice of lower sets of a finite poset: its sup-irreducible elements form a poset P whose lower sets reconstruct the lattice, and every finite poset arises this way from its own lattice of lower sets.*

Let P be the finite poset of sup-irreducible elements of a finite frame. Then, by Birkhoff duality (OrderIso.lowerSetSupIrred), every finite frame is the

lattice of *lower sets* of P . We fix the following notation, used throughout the rest of the paper. A *lower set* $L \subseteq P$ satisfies $p \in L, q \leq p \Rightarrow q \in L$, and an *upper set* is defined dually. Lower sets are always denoted L , and upper sets U (opens of a topological space are still U, V, W , as in traditional usage). The *principal lower set* at p is $\downarrow p = \{q : q \leq p\}$, the *strict principal lower set* at p is $\Downarrow p = \{q : q < p\}$, and the *principal upper set* at p is $\uparrow p = \{q : q \geq p\}$. We prove the finite representation theorem below on the lower sets of P . Transporting it across the isomorphism above then gives the theorem for every finite frame, not just lower sets^{Lean}. Let us define the *mass* of a point as the discrete derivative of v^{Lean} :

$$\text{mass}(p) = v(\downarrow p) - v(\Downarrow p).$$

The masses recover v and sum to 1:

Theorem 15 (finite representation^{Lean}). *For a finite P , any valuation v on $\text{Low}(P)$, and any $L \in \text{Low}(P)$, $v(L) = \sum_{p \in L} \text{mass}(p)$, and $\sum_p \text{mass}(p) = 1$.*

The proof works by induction on $|L|$: repeatedly remove a maximal point p from L , and use modularity applied to $(L \setminus \uparrow p, \downarrow p)$ to show each removal changes v by exactly $\text{mass}(p)$. The mass function is a classical probability mass function on P^{Lean} . And since lower sets are exactly the opens of the Alexandrov topology, where interior is the identity, the theorem also reads $v = \mu \circ \text{int}$, i.e. every valuation on a finite frame is the classical measure μ applied after taking the interior. In the finite case, this is the converse of the measure-theoretic bridge of Section 7. The intuitionistic mass also agrees with the classical notion of the mass of a point: reading $v = \mu \circ \text{int}$,

$$\text{mass}(p) = \mu(\downarrow p) - \mu(\Downarrow p) = \mu(\{p\}),$$

since $\downarrow p$ is the disjoint union of $\Downarrow p$ and $\{p\}$ and μ is defined on all subsets of P .

Theorem 15 represented v as a sum of point masses. The dual view represents v as a mixture of sharp *point-valuations*. For each $p \in P$, let δ_p be the valuation with $\delta_p(L) = 1$ if $p \in L$, and $\delta_p(L) = 0$ otherwise:

Theorem 16 (mixture characterization^{Lean}). *On a finite frame, every valuation v satisfies $v = \sum_p \text{mass}(p) \cdot \delta_p$.*

Together with the fact that any mixture of valuations is again a valuation^{Lean}, this theorem characterizes the valuations on a finite frame exactly: they are the finite convex mixtures of point-valuations, closed under mixing because modularity is linear. Section 6 shows modularity cannot instead be *derived* from more primitive axioms. It must be posited outright, and doing so is exactly what makes every valuation a mixture of sharp, all-or-nothing certainties.

4.2 Infinite Boundary Representation via Scott Continuity

Both halves of the finite theorem fail for infinite frames:

Example 17 (obstruction^{Lean}). On $\text{Low}(\mathbb{N})$, the indicator v of $\top$ is a valuation with $\text{mass}(p) = 0$ for every point p , and $v(\top) = 1$.

No point carries any of the mass. The failure is one of approximation: $v(\top)$ is not the supremum of $v(\downarrow n)$ over the finite stages. In order to recover the representation, we introduce Scott continuity:

Definition 18 (directed set, Scott continuity). A subset $D \subseteq \Omega$ is *directed* if, for any finite subset $F \subseteq D$, there exists an element $d \in D$ that is an upper bound of F (taking $F = \emptyset$ shows a directed set is nonempty). A valuation v is *Scott-continuous* if $v(\bigsqcup D) = \sup_{a \in D} v(a)$ for every directed D .

The $\top$ -indicator of Example 17 is not Scott-continuous^{Lean}, and Scott continuity is exactly what restores the representation:

Theorem 19 (recovery^{Lean}). Let v be a valuation on $\text{Low}(\mathbb{N})$. If $v(\top) = \sup_n v(\downarrow n)$ then $v(\top) = \sum_n \text{mass}(n)$.

In general, on any poset all of whose principal lower sets are finite, Scott continuity implies zero diffuse part without additional preconditions. However, subtraction in $[0, \infty]$ is truncated at 0, so if $\sum_p \text{mass}(p)$ were greater than $v(\top)$, the expression $v(\top) - \sum_p \text{mass}(p)$ would still evaluate to 0. Theorem 20 rules out $\sum_p \text{mass}(p) > v(\top)$ unconditionally, so the subtraction below is never truncated:

Theorem 20 (general decomposition^{Lean}). For every valuation v on $\text{Low}(P)$ and an arbitrary poset P , $\sum_p \text{mass}(p) \leq v(\top)$.

By Theorem 20 the remainder $v(\top) - \sum_p \text{mass}(p)$ is a genuine difference. Every valuation thus splits unconditionally into an *atomic* part carried by points ($\sum_p \text{mass}(p)$, which is always a sub-probability), and a *diffuse* (nonnegative) remainder $v(\top) - \sum_p \text{mass}(p)$. This is a constructive Lebesgue-style decomposition. The finite and Scott cases are the purely atomic extremes, and the $\top$ -indicator is the purely diffuse one. The proof of Theorem 20 re-runs the maximal-point induction on each finite subset of P , bounding its partial sum by $v(\top)$, then takes the supremum over all finite subsets to bound $\sum_p \text{mass}(p)$ itself. Its proof does not directly invoke an excluded-middle case split. However, it relies on the (classical) decidability of the ordering of the reals (a default in mathlib). Representing the diffuse part on non-spatial frames remains the open frontier. The obstruction there is again that the frame has *too few points*, the same one as behind the undecidability of Section 7.

5 The Dempster–Shafer Reading

Sections 2–4 independently arrive at the same structure as Dempster–Shafer (DS) theory [Dempster 1967; Shafer 1976], including sub-additive complements, a belief/plausibility interval, and mass functions. This section makes the identification exact in both statics and dynamics. In each case, it

locates the precise point where excluded middle makes a difference.

5.1 Total Monotonicity via Constructive Inclusion–Exclusion

The $\neg\neg$ -stable (*regular*) elements of a frame form a Boolean algebra with meet $\sqcap$ and join $\neg\neg(a \sqcup b)$ (Glivenko). Let us call the resulting algebra the *Booleanization* of this set.

Definition 21 (belief function, plausibility function). Let B be a Boolean algebra. A DS *belief function* on B is a monotone map $\text{Bel} : B \rightarrow [0, 1]$ with $\text{Bel}(\perp) = 0$ and $\text{Bel}(\top) = 1$ that is *n-monotone* for every $n \geq 2$: for all $A_1, \dots, A_n \in B$,

$$\text{Bel}\left(\bigvee_{i=1}^n A_i\right) \geq \sum_{\emptyset \neq T \subseteq \{1, \dots, n\}} (-1)^{|T|+1} \text{Bel}\left(\bigwedge_{i \in T} A_i\right).$$

Its *plausibility function* is the map $\text{Pl} : B \rightarrow [0, 1]$ given by $\text{Pl}(A) = 1 - \text{Bel}(\neg A)$.

We show v , restricted to its Booleanization, is such a belief function. The 2-monotone case is immediate from modularity^{Lean}. The full tower follows from the following result:

Theorem 22 (inclusion–exclusion^{Lean}). For every valuation on every frame Ω , every finite family $a : \iota \rightarrow \Omega$, and every finite $s \subseteq \iota$, inclusion–exclusion holds with equality for the frame join:

$$v\left(\bigsqcup_{i \in s} a_i\right) = \sum_{\emptyset \neq T \subseteq s} (-1)^{|T|+1} v\left(\bigsqcap_{i \in T} a_i\right).$$

We state this in Lean additively, with the odd-cardinality half of the signed sum set against the even-cardinality half, so that no $[0, \infty]$ subtraction occurs. The proof is by induction with modularity and frame distributivity.

Replacing the frame join $\bigsqcup_i a_i$ on the equality’s other side with its double negation $\neg\neg(\bigsqcup_i a_i)$, the Booleanization’s own join, yields the following inequality:

Theorem 23 (total monotonicity^{Lean}). For every valuation on every frame Ω , every finite family $a : \iota \rightarrow \Omega$, and every finite $s \subseteq \iota$:

$$\sum_{\emptyset \neq T \subseteq s} (-1)^{|T|+1} v\left(\bigsqcap_{i \in T} a_i\right) \leq v\left(\neg\neg\left(\bigsqcup_{i \in s} a_i\right)\right).$$

The Booleanization’s join is given by $\neg\neg a \sqcup$, and its meets of regular elements are again regular. Interpreted in this way, Theorem 23 states exactly the ∞ -monotone tower: v restricted to its Booleanization is a belief function.

The proof of Theorem 23 from Theorem 22 follows from the monotonicity along $a \leq \neg\neg a$. Here, inclusion–exclusion does *not* fail constructively. It is exact for the intuitionistic disjunction. Applying the double-negation nucleus (a closure operator preserving finite meets) $\neg\neg$ to the join on the equality’s right-hand side can only enlarge it. This follows from the fact that $a \leq \neg\neg a$ and v are monotone, which turns the equality into the inequality of Theorem 23. The

amount by which the right-hand side is enlarged is exactly the double-negation gap of Section 2.

Definition 21 is the classical notion, stated on an abstract Boolean algebra. In our setting each of its ingredients is already present: the Boolean algebra is the Booleanization, the belief function is v itself, restricted to the regular elements and evaluated against the Boolean join $\neg\neg\circ\sqcup$, and the plausibility function is $\text{Pl}(a) = 1 - v(\neg a)$. Additive probability does not become a belief function by weakening an axiom, since modularity for v is never relaxed. What changes is only the join used to state the tower. The belief–plausibility interval is governed by the slack:

Theorem 24 (interval width^{Lean}). *For every valuation v on a frame Ω and every $a \in \Omega$, $\text{Pl}(a) - v(a) = \text{slack}(a)$.*

5.2 The Conditioning Characterization

DS theory carries two update rules that coincide classically. *Geometric* conditioning renormalizes belief, and under the DS dictionary it is the localic posterior $v(a \sqcap b)/v(b)$ of Section 2.

Definition 25 (Dempster’s rule of conditioning^{Lean}). For a valuation v on a frame Ω , elements $a, b \in \Omega$, and evidence of nonzero plausibility, $v(b) \neq 0$, *Dempster’s rule of conditioning* [Shafer 1976] states that

$$v(a \parallel b) = \frac{v(a \sqcup \neg b) - v(\neg b)}{1 - v(\neg b)}.$$

The hypothesis $v(b) \neq 0$ makes the denominator nonzero, since $v(b) + v(\neg b) \leq 1$ gives $v(b) \leq 1 - v(\neg b)$.

That is, the DS conditioning rule discards the mass incompatible with the evidence and renormalizes by its plausibility. Both the geometric posterior of Section 2 and Dempster’s rule of Definition 25 are again valuations. Their modularity is the frame distributivity $(a \sqcup \neg b) \sqcap (a' \sqcup \neg b) = (a \sqcap a') \sqcup \neg b$. Their disagreement is given by the *excluded-middle gap* $\text{emGap}(a, b) = v(a) - v(a \sqcap (b \sqcup \neg b))$, i.e. the mass of a outside b ’s instance of excluded middle:

Lemma 26 (the join–complement identity^{Lean}). *For every valuation v on every frame Ω and every $a, b \in \Omega$, $v(a \sqcup \neg b) = v(\neg b) + v(a \sqcap b) + \text{emGap}(a, b)$.*

Dempster’s numerator therefore exceeds Bayes’s by exactly $\text{emGap}(a, b)$. This gap depends on a , but whether it vanishes at every a depends only on the slack at the evidence b :

Theorem 27 (the conditioning characterization^{Lean}). *For every valuation v on every frame Ω and every $b \in \Omega$ with $v(b) \neq 0$, $(v(a \parallel b) = v(a \sqcap b)/v(b) \text{ for every } a \in \Omega) \text{ iff } \text{slack}(b) = 0$.*

The hypothesis $\text{slack}(b) = 0$ eliminates the gap $\text{emGap}(a, b)$ for every a and makes the two normalizers coincide, since $1 - v(\neg b) = v(b)$. Conversely, testing at

$a := b$, geometric conditioning gives certainty 1, while Dempster gives $v(b)/(1 - v(\neg b))$, which is 1 only if the slack vanishes. Theorem 27 is the valuation-relative analogue of Theorem 10: both are iff statements about the same proposition, $b \sqcup \neg b = \top$.

We call the frame-level fact *static* and the valuation-level fact *dynamic*: the former is an equation in the frame itself, between $b \sqcup \neg b$ and $\top$, with no valuation involved, while the latter tests the same proposition through v , since $\text{slack}(b) = 0$ is exactly $v(b \sqcup \neg b) = v(\top)$. Dempster and Bayes therefore coincide at b whenever v cannot distinguish $b \sqcup \neg b$ from $\top$, even if the two differ in the frame itself. In the measure model, Dempster and Bayes disagree precisely when the evidence’s boundary carries mass.

6 Cox’s Theorem

Cox’s theorem [Cox 1946; Jaynes 2003] derives the probability calculus from conditions on plausibility. After Halpern’s counterexample [Halpern 1999], its repaired forms [Arnborg and Sjödin 2000; Horn 2003] fix the calculus up to monotone regreduation. Every known derivation is classical, and Section 3 showed why: the negation axiom R3 is excluded middle. Dropping R3 leaves the product rule unaffected, since it is logic-free, and requires the sum rule to become an axiom rather than a theorem. Separately, the naive constructive statement, demanding a regreduation strictly monotone on all of $\mathbb{R}$, is unsatisfiable (Section 6.3).

6.1 A Cox Structure without Negation

Definition 28 (Cox model^{Lean}). A *Cox model* on a frame Ω is a plausibility assignment $\text{pl} : \Omega \times \Omega \rightarrow \mathbb{R}$, written $\text{pl}(a, b)$ for the plausibility of a given b , together with a *conjunction functional* $F : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$, such that $\text{pl}(\cdot, c)$ is monotone for every c , the boundary conditions $\text{pl}(\perp, c) = 0$ and $\text{pl}(\top, c) = 1$ hold for every c , the product rule $\text{pl}(a \sqcap b, c) = F(\text{pl}(a, b \sqcap c), \text{pl}(b, c))$ holds, F is associative, and $x \mapsto F(x, y)$ is strictly monotone for every $y > 0$.

The missing R3 axiom can be observed in the omission of any constraint on the relation between $\text{pl}(\neg a, b)$ and $\text{pl}(a, b)$. The definition restricts the monotonicity of F to $0 < y$ to avoid a contradiction with the fact that the boundary axioms force $F(x, 0) = 0$ for all x . An explicit model on the non-classical chain $[0, \infty]$ satisfies this constraint^{Lean}, as $a = \top$ is provably decidable.

6.2 The Sum Rule Is Irreducible

Classically, additivity is *derived*: R3 turns disjunction into negated conjunction via De Morgan, and the product rule does the rest. Constructively that route is closed, since $a \vee b \neq \neg(\neg a \wedge \neg b)$ in general:

Theorem 29 (the sum rule is irreducible^{Lean}). *Let Ω be the five-element frame of lower sets of the poset $o < a, o < b$ with a, b incomparable. Then there is a monotone, normalized*

assignment q on Ω that is additive on disjoint joins yet not modular, and no binary operation S on values satisfies $q(x \sqcup y) = S(q(x), q(y))$ for all x, y . Moreover, the pairs $(\downarrow a, \downarrow a)$ and $(\downarrow a, \downarrow b)$ have identical value-pairs $(\frac{1}{2}, \frac{1}{2})$ but joins valued $\frac{1}{2}$ and 1.

Unlike for a conjunction, a disjunction's plausibility cannot generally be expressed as its disjuncts' plausibilities via conditioning. So, a constructive Cox theorem *must* posit the sum-rule half. Modularity, which says that the calculus is closed under mixing certainties, is what posits this, justified by Theorem 16. The failure is not repaired by strengthening the logic short of Boolean: the counterexample frame satisfies weak excluded middle^{Lean}, since every nonempty lower set contains the bottom point o , so the irreducibility persists under that intermediate assumption.

6.3 Cox Model Theorem

A regratuation is a map $g : \mathbb{R} \rightarrow [0, \infty]$ with $g(0) = 0$. No such map is strictly monotone on all of $\mathbb{R}$, since strictness would force $g(-1) < g(0) = 0$, impossible in $[0, \infty]$. The repair (strictness only on $[0, 1]$, where plausibilities live) is exactly the same domain restriction that Halpern's critique of the classical theorem turns on. The corrected statement is:

Theorem 30 (the constructive Cox theorem^{Lean}). *For every frame Ω , every Cox model on Ω whose unconditional plausibility is modular regratuates to a valuation: there exist $v : \text{Valuation } \Omega$ and g strictly monotone on $[0, 1]$ with $g(0) = 0, g(1) = 1$, and for all $a \in \Omega, v(a) = g(\text{pl}(a, \top))$.*

The regratuation for the sum-rule half is straightforward ($g = \text{ENNReal.ofReal}$). The product-rule half, showing F regratuates to multiplication, rests on Aczél's associativity theorem [Aczél 1966], a statement of pure analysis whose type never mentions Ω . mathlib has no Aczél theorem, no Hölder-style embedding for ordered semigroups on intervals, and no Cauchy functional equation under monotonicity, so we build the analytic core from scratch, obtaining a normalized, strictly monotone multiplicative generator on the positive cone^{Lean}. The construction also weakens Hölder's classical hypotheses [Hölder 1901]: the Archimedean property is *derived* from continuity and positivity rather than assumed, and the interval form of divisibility follows from the intermediate value theorem^{Lean} applied to a plain divisibility hypothesis^{Lean}. The constructive/classical divide is therefore *entirely* in the sum rule: nothing in the product-rule apparatus mentions Ω . Appendix C gives the construction. The residual analytic gap is discussed in Section 9.

7 Two Grounding Models

Two concrete models demonstrate that our abstract theory is non-degenerate. One shows every classical measure *contains*

an intuitionistic probability. The other shows the intuitionistic theory cannot collapse back without solving the halting problem.

7.1 Measures through the Interior Operator

Gödel–McKinsey–Tarski interpret intuitionistic logic in modal S4: propositions are *open* sets, and the modal necessity operator $\Box$ of S4 is interpreted as the topological interior, written $\text{int}(\cdot)$ below [Gödel 1933; McKinsey and Tarski 1944]. Our valuations quantify that same translation:

Theorem 31 (measures as valuations^{Lean}). *Let μ be a probability measure on a space $X, A \subseteq X$, and define $v(A) = \mu(\text{int } A)$. Since $\text{int } U = U$ for open U , v restricts on the frame $O(X)$ to $U \mapsto \mu(U)$, and this restriction is a valuation.*

Modularity is $\mu(U \cup V) + \mu(U \cap V) = \mu(U) + \mu(V)$. The slack of an open U is then $1 - \mu(U) - \mu(\text{int } U^c)$, which is the measure of the boundary ∂U . We can now prove that, in general, slack is exactly the boundary mass:

Theorem 32 (slack is boundary mass^{Lean}). *For any probability measure μ on a topological space X and any open U , $\text{slack}(U) = \mu(\partial U)$.*

The Heyting complement of U in $O(X)$ is the interior of its set complement. So, the undecided region, the complement of $U \sqcup \neg U$, is exactly $\overline{U} \setminus U = \partial U$. The general sub-additivity inequality of Section 2 still holds here^{Lean}. This identification of slack with boundary mass also exhibits the modal identification $\text{Bel} = P(\Box \cdot)$ of the DS literature [Ruspini 1986] arising from the logic itself. Theorem 15 is the converse in the finite case (in the general case, the converse is the open problem, see Section 4).

To characterize weak excluded middle spaces, we give the (standard) definitions required:

Definition 33 (extremally disconnected). A topological space X is *extremally disconnected* if the closure of every open set is again open.

Definition 34 (Stone space, Stonean space). The *Stone space* of a Boolean algebra B is the space of ultrafilters on B (equivalently, the 2-valued homomorphisms $B \rightarrow \{0, 1\}$), topologized by the clopen basis $\{U : b \in U\}$ for $b \in B$. *Stone duality* [Johnstone 1982] identifies Boolean algebras with exactly the compact, Hausdorff, totally disconnected spaces arising this way. A *Stonean space* is a compact Hausdorff space that is also extremally disconnected. Equivalently, it is the Stone space of a *complete* Boolean algebra.

Remark 35 (weak excluded middle on opens^{Lean}). Section 2 noted that imposing weak excluded middle on the whole frame collapses slack to pure double-negation instability. For $\Omega = O(X)$, this global condition holds exactly when X is extremally disconnected.

An immediate corollary gives the characterization:

Corollary 36 (Stonean characterization^{Lean}). *Given a compact Hausdorff space X , weak excluded middle on $O(X)$ holds exactly when X is a Stonean space.*

7.2 Slack Does Not Determine the Valuation

Submodularity (Theorem 7) is a necessary condition on slack. It is not sufficient: distinct valuations, each satisfying every axiom a valuation must satisfy, can share a slack function.

Example 37 (boundaries in the three-point space^{Lean}). Let $\text{ThreePt} = \{p_A, p_B, p_Z\}$ carry the topology generated by the two singletons $\{p_A\}$ and $\{p_B\}$, so its opens are exactly $\emptyset$, $\{p_A\}$, $\{p_B\}$, $\{p_A, p_B\}$, and the whole space. For every open U , $\partial U \subseteq \{p_Z\}$.

This is a fact about the topology alone, independent of any measure: p_Z has no proper open neighbourhood, so it is the only point any open's frontier can contain.

Theorem 38 (non-recovery^{Lean}). *There exist valuations $v \neq v'$ on $O(\text{ThreePt})$ with $\text{slack}(v) = \text{slack}(v')$ as functions.*

For weights $a, b, c \geq 0$ with $a + b + c = 1$, the point measure $\mu_{a,b,c} = a \delta_{p_A} + b \delta_{p_B} + c \delta_{p_Z}$, read through Section 7.1's bridge, gives a valuation $v_{a,b,c}$ on $O(\text{ThreePt})$. By Example 37 and Theorem 32, its slack at an open U is $\mu_{a,b,c}(\partial U)$, equal to c when $p_Z \in \partial U$ and to 0 otherwise: a quantity depending only on c , not on how the remaining mass splits between a and b . So $v_{1/4,1/4,1/2}$ and $v_{1/8,3/8,1/2}$ have identical slack functions while disagreeing at $\{p_A\}$, where they take the values $1/4$ and $1/8$. Both are fully modular, monotone, and normalized: nothing in Theorem 7 or its proof distinguishes them. Weak excluded middle does not rescue recoverability. If anything, the fully classical case is the worst one for it. When X is extremally disconnected, the De Morgan gap vanishes (Section 7.1) and slack degenerates to the double-negation gap alone, but on a space where excluded middle holds outright that gap vanishes too, so slack is identically 0 for every valuation. The zero slack function is then shared by every probability measure on X , which is the most extreme failure of recovery available.

7.3 The Computability Guard

The theory developed here is non-degenerate only if there are spaces whose slack does not vanish identically. Computability supplies them:

Definition 39 (Sierpiński topology, semidecidable proposition). The *Sierpiński space* $\mathbb{S}$ is the type Prop of propositions carrying the topology with exactly three opens: $\emptyset$, the set $\text{halts} = \{q \mid q\}$ of true propositions, and $\mathbb{S}$ itself [Escardó 2004; Rosolini 1986]. A proposition is *semidecidable* if it is an open in this topology.

The topology encodes verifiability. A point of $\mathbb{S}$ is a proposition, say “this computation halts.” The one nontrivial observation available is membership in halts , established by

producing a proof (running the computation to termination). No open captures the complementary observation, and on the frame $O(\mathbb{S}) = \{\perp, \text{halts}, \top\}$ that Σ_1 asymmetry is a theorem:

Theorem 40 (the Σ_1 asymmetry^{Lean}). $\neg \text{halts} = \perp$.

Every nonempty open of $\mathbb{S}$ contains True , so $\perp$ is the only open disjoint from halts . Thus, no nonempty open of $\mathbb{S}$ is evidence that the computation does not halt.

We now instantiate the measure-to-valuation construction of Theorem 31 at $X = \mathbb{S}$, allowing us to compute the slack:

Example 41 (halting slack^{Lean}). Let $\mu = p \cdot \delta_{\text{True}} + (1-p) \cdot \delta_{\text{False}}$, and let v be the resulting valuation. Then

$$\text{slack}(\text{halts}) = 1 - p = \mu(\partial \{q \mid q\}) = \mu\{\text{False}\}.$$

Since halts is already open, $v(\text{halts}) = \mu(\text{halts}) = p$. The DS “ignorance” of Section 5 is here the mass of the single point False , the one outcome no finite observation can ever resolve. This probability is a fact about the model, not about anyone's ignorance.

Chaitin's Ω . The same bridge scales to algorithmic information theory. Fix a universal prefix-free machine U , *prefix-free* in that no halting program of U is a proper prefix of another, and *universal* in that it simulates every other prefix-free machine. Give 2^ω , the space of infinite bit strings, the fair-coin measure μ . The event H of having a halting program of U as a prefix is open, since halting is established by a finite prefix (Definition 39), and prefix-freeness makes H a disjoint union of cylinders, so $v(H) = \mu(H) = \sum_{p \text{ halts}} 2^{-|p|}$: exactly Chaitin's halting probability Ω [Chaitin 1975]. Its slack $\mu(\partial H)$ is the mass of strings approximated arbitrarily well by halting strings while never having a halting program as a prefix, and Ω 's classical uncomputability is the aggregate form of the pointwise obstruction above. The measure-theoretic core is mechanized on Cantor space with mathlib 's infinite product measure: basic cylinders are clopen with measure $2^{-|w|}$ ^{Lean}, and a *finite* prefix-free set of words has pairwise disjoint cylinders of total measure $\sum_w 2^{-|w|}$ ^{Lean}, the Kraft-inequality content underlying Ω . The universal machine itself, and the countable extension of the finite Kraft sum, are computability theory beyond our scope.

Definition 42 (halting problem). For a code c and input n , the *halting problem* is the predicate $H(c, n) = \text{“the machine with code } c \text{ halts on input } n\text{”}$ (ComputablePred).

Slack cannot be eliminated by simply declaring a classical answer for each code. The family of valuations defined below attempts to do this:

Definition 43 (sharp classical readout). For a given n , the *sharp classical readout* is the family of slack-free valuations $\rho_n(c) \in \text{Valuation}(O(\mathbb{S}))$, with $\rho_n(c)(\text{halts}) = 1$ if $H(c, n)$ holds, and $\rho_n(c)(\text{halts}) = 0$ otherwise.

However, this family is provably not computable:

Theorem 44 (the sharp readout is uncomputable^{Lean}). *For every n , the predicate $c \mapsto (\rho_n(c)(\text{halts}) = 1)$ is not computable.*

The proof identifies the predicate with $H(\cdot, n)$ and applies the undecidability of the halting problem (ComputablePred.halting_problem).

We can conclude that slack is not a defect of the constructive theory: any axiom strong enough to force classical collapse demands that its models compute the uncomputable. For the sum rule, this guard plays the role the explicit Cox-model instance plays for the product rule (Section 6): each axiom set in the paper comes with non-trivial models satisfying it.

8 Related Work

Informal precedents. Pairing probability with intuitionistic logic is not a new idea, though the literature on it is scattered. Weatherston develops intuitionistic probability axiomatically and defends it with Dutch-book arguments [Weatherston 2003]. His axioms match our valuation axioms exactly, with negation derived rather than assumed, so the object itself and the bound $v(a) + v(\neg a) \leq 1$ are his. What is new here is the decomposition of that bound's slack into dnGap and deMorganGap and its determinacy from v alone (Theorem 6). Paris gives an informal theorem that any modular assignment of non-classical truth values is exactly a Dempster–Shafer belief function [Paris 1994]. We mechanize the analogous fact for our intuitionistic setting: the 2-monotone case follows directly from modularity^{Lean}, and the full tower is Theorem 23. The identification $\text{Bel} = P(\Box \cdot)$ connecting belief functions [Dempster 1967; Shafer 1976] to modal logic is established [Ruspini 1986]. Section 7 shows this identification is not a separate axiom for us but a special case of our frame-level slack, recovered automatically once the frame is a topology and $\neg$ is read as the interior of the complement^{Lean}. On the localic side, valuations on frames are standard objects of constructive measure theory, with Vickers's work and the Coquand–Spitters development as canonical references [Coquand and Spitters 2009; Vickers 2008]. These normally build in Scott continuity and target additive, Riesz-style goals. Section 4 shows Scott continuity to be a substantive assumption (it decides point-representability on the frames we examine), so our base structure omits it. Cox, Jaynes, Halpern's counterexample, and the Van Horn and Arnborg–Sjödén repairs [Arnborg and Sjödén 2000; Cox 1946; Halpern 1999; Horn 2003; Jaynes 2003] are classical throughout, and Cox's and Van Horn's derivations both lean on double-negation elimination. Section 6 proves the sum-rule half without that move, at the price of positing modularity as an axiom rather than deriving it (Theorem 29). Van Horn observes that dropping axiom R3 yields an interval-valued theory, exactly the belief/plausibility pair our slack separates. To

our knowledge, no prior work, informal or formal, states either characterization theorem: complement rule $\iff$ excluded middle (Theorem 10), or Dempster = Bayes $\iff$ zero slack at the evidence (Theorem 27).

Prior formalizations. Two bodies of mechanized work are adjacent, and neither overlaps. ALEA [Audebaud and Paulin-Mohring 2009] formalizes a discrete distribution monad in Coq for randomized programs: additive probability, no event logic in our sense. Closest is the HoTT/Coq line of Faissole and Spitters on *valuations* over lower reals [Faissole and Spitters 2017], done constructively with Scott continuity and aimed at integration (Riesz) and a Giry-style valuation *monad*. Their valuations differ from ours in carrying Scott continuity, and their monad is the Scott-continuous instance of the question examined in Section 9. Our development drops the continuity their base structure builds in, and the two mechanizations do not otherwise overlap.

Non-Boolean probability elsewhere. Quantum probability on orthomodular lattices, with Gleason's theorem [Birkhoff and von Neumann 1936; Gleason 1957], is the long-standing precedent for re-basing probability on a non-Boolean event logic. Orthomodular lattices give up distributivity itself, while our frames stay distributive by definition (Definition 1), locating all the non-classical behavior in the failure of excluded middle alone. Free probability, fuzzy measures, and imprecise probability [Walley 1991] vary other parameters (independence, additivity, single-valuedness). Where Walley's coherent lower previsions posit the belief/plausibility gap directly as a coherence primitive, our slack (Section 2) is derived, forced by modularity and negation rather than assumed. Fritz's synthetic Markov-category framework [Fritz 2020] varies the ambient structure categorically rather than order-theoretically, axiomatizing conditional independence and sufficient statistics abstractly across discrete, measure-theoretic, and Gaussian probability alike. It remains unmechanized and is orthogonal to our variation in the underlying logic. Wolpert and Kinney model mathematics itself as a stochastic process, replacing binary theoremhood with a probability distribution [Wolpert and Kinney 2024]. Theoremhood is a semidecidable event, so the computability guard of Section 7 bears directly on that program: slack-free credences about such events are uncomputable. Our variation is specifically the *logic*, holding the Cox-style methodology fixed. Within DS theory, the two conditionings and their inequivalence are classical [Fagin and Halpern 1991; Shafer 1976]. What is new in Theorem 27 is the exact logical characterization of their agreement, and its mechanization.

9 Conclusion

Many properties of programs are only semidecidable, so the Boolean event algebra Kolmogorov's axioms [Kolmogorov

1933] presuppose is not a good fit for reasoning about them. The notion of a degree of certainty does not presuppose Boolean logic. Classicality enters only through the complement rule (Theorem 10). Its dynamic counterpart is the agreement of Dempster and Bayesian conditioning (Theorem 27). Everything else holds verbatim on frames: conditioning, Bayes symmetry, disjoint additivity, total probability over genuine partitions, and inclusion–exclusion itself (Theorem 22). The weaker logic adds the slack structure, along with its two-gap decomposition, and the atomic/diffuse divide. For this new structure, we demonstrate a belief-function reading through $\neg\neg$, and give a computability-theoretic reason that structure cannot be dispensed with: forcing it away on a semidecidable event is exactly as hard as solving the halting problem (Section 7).

Functoriality and the monad question. For a fixed choice of logic, the connection to the categorical picture is concrete. Pushforward along a frame homomorphism $f : \Omega \rightarrow \Omega'$ sends a valuation v on Ω' to $(f_*v)(a) = v(f(a))$ on Ω , and this preserves all four valuation axioms, making $\text{Val}(-)$ a functor on locales, the localic counterpart of the functor part of the Giry monad [Giry 1982]. This is mechanized end to end, the pushforward together with the two functor laws checked against identity and composition of frame homomorphisms^{Lean}. The point valuations δ_p of Section 4 are the unit, already formalized, and Theorem 15 shows that on finite frames Val agrees, on objects, with the classical finite-distribution monad on points. The multiplication is where the substance is. Flattening a valuation of valuations is integration, hence product valuations and Fubini, and by Kock’s identification of commutative monads with Fubini theorems [Kock 2012], whether statistics-as-kernel-composition transports to a weaker logic is precisely the question whether that logic’s valuation functor is a commutative monad. Two machine-checked results delimit the multiplication from both sides.

On the chain product, the question of *which* product valuation to take does not arise, since the frame $\text{Low}(\mathbb{N} \times \mathbb{N})$ is *rigid*: every lower set is a *finite* union of rectangles, because beyond a stabilization height every occupied column is full, so the full columns and finitely many rows account for everything^{Lean}. The rigidity mechanism is general. By the inclusion–exclusion equality, two valuations agreeing on any $\sqcap$ -closed family agree on all finite joins of its members^{Lean}, and the chain product is the instance where that family is the rectangles. The product valuation there, whenever it exists, is therefore unique, with no continuity assumed^{Lean}.

On the powerset frame $\text{Set}(\mathbb{N} \times \mathbb{N})$, whose elements are far from finite unions of rectangles, non-uniqueness is a theorem. The indicator of an ultrafilter is a valuation^{Lean}, and any ultrafilter on a product splits rectangles through its marginal pushforwards^{Lean}. Taking a free ultrafilter $\mathcal{U}$ on $\mathbb{N}$, the two iteration orders of its tensor with itself and its diagonal

pushforward carry identical values on *every* rectangle, yet they differ on the upper triangle $\{(m, m') \mid m < m'\}$ ^{Lean}. The two orders of iterated integration of the same marginal data therefore disagree: Fubini fails, concretely^{Lean}. Compare the Fubini pathologies of non-additive integration [Denneberg 1994; Ghirardato 1997]. By Kock’s correspondence, no canonical commutative multiplication exists for unrestricted valuations. On the powerset frame these valuations are finitely additive probability charges in the sense of [Rao and Rao 1983], so this half of the dichotomy is not specific to intuitionism: the freedom is already present in the classical finitely additive setting, and dropping continuity is what admits it. A monad therefore requires restricting either the frames or the valuations.

The obstruction to a monad. The question upstream of the multiplication is worth stating precisely, though what follows is an assessment rather than a theorem. A monad needs a category for Val to be an endofunctor on. The space of valuations must itself be an object, a “space” whose opens one can speak of. For Scott-continuous valuations that space is a locale and monad structure is available, constructively, through the probabilistic powerdomain line [Coquand and Spitters 2009; Jones and Plotkin 1989; Vickers 2008]. Even there, exhibiting a cartesian closed category closed under the construction is the Jung–Tix problem, open since 1998 for the classical probabilistic powerdomain functor itself [Jung and Tix 1998], though reframed versions of the category have recently been settled. Dropping continuity makes the difficulty prior rather than merely worse, since the standard presentations of the valuation space use continuity essentially.

One candidate route to a well-posed question does exist. The localic treatments present the space of continuous valuations as the classifying locale of a propositional geometric theory [Coquand and Spitters 2009; Vickers 2008], with generators asserting $q < v(a)$ for rational q and axioms for monotonicity, normalization, modularity (expressible with rational parameters and countable disjunctions), and continuity. Deleting the continuity axiom leaves a geometric theory, so a classifying locale still exists on general grounds, and classically its points are exactly the valuations of this paper. We have verified none of the structure this candidate would need. The unit looks unproblematic, but the multiplication requires integrating the evaluation maps against a merely modular, discontinuous valuation, where the sum of two lower-real-valued maps is an infinite join, and the diffuse part of Example 17 is precisely what such joins fail to capture. We offer the formulation as a target, not a result.

A more modest strategy, with a better track record, is to carve out a fragment rather than construct the whole monad. On the measure-theoretic side, the failure of cartesian closure for measurable spaces was answered by quasi-Borel spaces [Heunen et al. 2017]. On the domain-theoretic side,

commutativity of the full valuations monad on DCPO is open, and the response is a hierarchy of commutative *submonads* generated from simple valuations [Jia et al. 2021]. Our decomposition theorem names the corresponding fragment here: the atomic part. As a first machine-checked step in that direction, we show that the countably-presented fragment composes correctly with no space structure needed at all. A countable mixture of valuations is a valuation^{Lean}, a mixture of mixtures flattens to a mixture with product weights, which is discrete Chapman–Kolmogorov^{Lean}, the one-point mixture is the identity^{Lean}, and iterated countable mixing is order-independent^{Lean}, all by the unconditional Fubini of $[0, \infty]$ sums over sigma types. This is associative, unital, commutative kernel composition for countably-atomic data, in direct contrast with the non-uniqueness above. It is not yet a monad: the presentations are not canonical, we do not quotient by “presents the same valuation,” and the diffuse part is exactly what no countable presentation captures.

In summary, the unit, the finite level, and countable kernel composition with its commutativity are proved. Rigidity on chain products and non-uniqueness on the powerset frame are proved, so the general multiplication is settled: it is not canonically determined. What remains is the ambient category, a formulation problem before it is anything else.

Open problems and future work. Four problems are open: (1) representing the diffuse part on non-spatial frames, (2) completing the Aczél development (continuity of the generator and its extension beyond the cone), where the classical analysis is known [Aczél 1966], (3) deriving modularity from a criterion more primitive than the mixture characterization (Theorem 16), knowing it cannot come from disjunction data alone (Theorem 29), and (4) a machine-checked Dutch-book theorem in the style of [Weatherson 2003]: the informal proof exists, and to our knowledge no proof assistant has one for *any* logic. Problems (1) and (3) are open mathematically, (2) and (4) mainly as formalizations. On the monad, two continuations are concrete: quotient the countable presentations into a Kleisli category of countably-atomic kernels, and mechanize the classifying-locale formulation above, which needs classifying locales of geometric theories, infrastructure mathlib currently lacks. Both are formalization tasks rather than mathematical gaps, and both are confined to a fragment, since the unrestricted multiplication is provably not canonical. Beyond intuitionistic logic there is no ambient category in which “logic $\mapsto$ its probability calculus” is a functor. What transfers is the method: state the classical desiderata, delete the axioms the target logic no longer licenses, and let a proof assistant check what survives, what splits, and what becomes unsatisfiable. Linear logic is the natural next case: its event algebras are quantales, a valuation there should grade the consumption of evidence, and the analogues of our representation theorems and of slack,

for the failure of contraction, are open [Danos and Ehrhard 2011].

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A Values in the Extended Reals

Valuations take values in the extended non-negative reals $[0, \infty]$, following the measure-theoretic convention of mathlib. As a result, sums indexed by an arbitrary family are unconditionally defined, with no summability hypothesis, and they commute over dependent sums: for a function f on a sigma type $\Sigma a, \beta(a)$, one has $\sum_p f(p) = \sum_a \sum_b f(a, b)$ with no side condition. This is what lets the countable-mixture results of Section 9 hold as stated. Additionally, subtraction is truncated at 0. This never affects the main text, since every subtraction there is applied to a pair already ordered by monotonicity, as with the mass of Section 4, so the truncation is never triggered.

B Lean Statements

B.1 Valuations on Frames

```

structure Valuation (Ω : Type*) [Order.
  Frame Ω] where
  toFun : Ω → ℝ≥0∞
  map_bot' : toFun ⊥ = 0
  map_top' : toFun ⊤ = 1
  mono' : Monotone toFun
  modular' : ∀ a b, toFun a + toFun b =
    toFun (a ⊔ b) + toFun (a ⊓ b)

theorem Valuation.add_compl_le_one (v :
  Valuation Ω) (a : Ω) :
  v a + v ac ≤ 1

theorem Valuation.add_compl_eq_sup (v :
  Valuation Ω) (a : Ω) :
  v a + v ac = v (a ⊔ ac)

theorem Valuation.le_one (v : Valuation Ω)
  (a : Ω) : v a ≤ 1

theorem Valuation.
  slack_eq_dnGap_add_deMorganGap (v :
  Valuation Ω) (a : Ω) :
  v.slack a = v.dnGap a + v.deMorganGap
    a

theorem Valuation.slack_eq_sub_add_sub (v
  : Valuation Ω) (a : Ω) :
  v.slack a = (v acc - v a) + (1 - (v a
    cc + v ac))

theorem Valuation.le_compl_compl (v :
  Valuation Ω) (a : Ω) : v a ≤ v acc

theorem Valuation.slack_submodular (v :
  Valuation Ω) (a b : Ω) :
  v.slack (a ⊔ b) + v.slack (a ⊓ b) ≤ v.
    slack a + v.slack b

def Order.Frame.IsDeMorgan (Ω : Type*) [
  Order.Frame Ω] : Prop :=
  ∀ a : Ω, acc ⊔ ac = ⊤

theorem Valuation.
  slack_eq_dnGap_of_isDeMorgan
  (hΩ : Order.Frame.IsDeMorgan Ω) (v :
  Valuation Ω) (a : Ω) :
  v.slack a = v.dnGap a

noncomputable def Valuation.condVal (v :
  Valuation Ω) (b : Ω) (hb : v b ≠ 0) :

```

```

1431     Valuation  $\Omega$ 
1432
1433   theorem Valuation.condVal_mul (v :
1434     Valuation  $\Omega$ ) (b :  $\Omega$ ) (hb : v b  $\neq$  0) (a
1435     :  $\Omega$ ) :
1436     v.condVal b hb a * v b = v (a  $\sqcap$  b)
1437
1438   theorem Valuation.condVal_symm (v :
1439     Valuation  $\Omega$ ) {a b :  $\Omega$ }
1440     (ha : v a  $\neq$  0) (hb : v b  $\neq$  0) :
1441     v.condVal b hb a * v b = v.condVal a
1442       ha b * v a
1443
1444   theorem Valuation.cond_add_compl_le (v :
1445     Valuation  $\Omega$ ) (a b :  $\Omega$ ) :
1446     v (a  $\sqcap$  b) + v (ac  $\sqcap$  b)  $\leq$  v b
1447
1448   theorem Valuation.
1449     add_compl_eq_one_of_complemented (v :
1450     Valuation  $\Omega$ ) {a :  $\Omega$ }
1451     (ha : a  $\sqcup$  ac =  $\top$ ) : v a + v ac = 1
1452
1453   variable [CompleteBooleanAlgebra  $\Omega$ ]
1454
1455   theorem classical_additivity (v :
1456     Valuation  $\Omega$ ) (a :  $\Omega$ ) :
1457     v a + v ac = 1
1458
1459   variable [CompleteBooleanAlgebra  $\Omega$ ]
1460
1461   theorem classical_slack_zero (v :
1462     Valuation  $\Omega$ ) (a :  $\Omega$ ) :
1463     v.slack a = 0
1464
1465   noncomputable def Valuation.mix (w :  $\iota \rightarrow \mathbb{R}$ 
1466      $\geq 0$ ) (hw :  $\sum i, w i = 1$ )
1467     (v :  $\iota \rightarrow$  Valuation  $\Omega$ ) : Valuation  $\Omega$ 
1468

```

B.2 Where Classicality Lives

```

1470
1471   def Valuation.HasClassicalNegation (v :
1472     Valuation  $\Omega$ ) : Prop :=
1473      $\forall a, v a + v a^c = 1$ 
1474
1475   theorem hasClassicalNegation_of_em (hem :
1476      $\forall a : \Omega, a \sqcup a^c = \top$ )
1477     (v : Valuation  $\Omega$ ) : v.
1478       HasClassicalNegation
1479
1480   theorem em_of_forall_hasClassicalNegation
1481     (h :  $\forall v : \text{Valuation } \Omega, v.$ 
1482       HasClassicalNegation) :  $\forall a : \Omega, a$ 
1483        $\sqcup a^c = \top$ 
1484

```

```

1486   theorem sharp_iff_point (v : Valuation  $\Omega$ )
1487     :
1488     v.IsSharp  $\leftrightarrow \exists (J : \text{Order.Ideal } \Omega) (hJ$ 
1489       : J.IsPrime) (htop :  $\top \notin J$ ),
1490       v = Ideal.toValuation J hJ htop
1491
1492   theorem
1493     sharp_scott_iff_completelyPrimePoint (v
1494       : Valuation  $\Omega$ ) :
1495     v.IsSharp  $\wedge v.$ ScottContinuous  $\leftrightarrow$ 
1496        $\exists (J : \text{Order.Ideal } \Omega) (hJ : \text{Ideal.}$ 
1497         IsCompletelyPrime J) (htop :  $\top \notin$ 
1498         J),
1499       v = Ideal.toValuation J hJ.1 htop
1500

```

B.3 Valuation Representation

```

1503   noncomputable def Valuation.mass (v :
1504     Valuation (LowerSet P)) (p : P) :  $\mathbb{R}_{\geq 0}$ 
1505
1506   theorem Valuation.eq_sum_mass (v :
1507     Valuation (LowerSet P)) (U : LowerSet P
1508     ) :
1509     v U =  $\sum p \in U.$ toFinset, v.mass p
1510
1511   theorem Valuation.sum_mass (v : Valuation
1512     (LowerSet P)) :  $\sum p, v.$ mass p = 1
1513
1514   noncomputable def Valuation.toPMF (v :
1515     Valuation (LowerSet P)) : PMF P
1516
1517   theorem Valuation.exists_pmf_presentation
1518     { $\Theta : \text{Type } u$ } [Order.Frame  $\Theta$ ] [Finite  $\Theta$ ]
1519     (v : Valuation  $\Theta$ ) :
1520      $\exists (P : \text{Type } u) (\_ : \text{PartialOrder } P) (\_$ 
1521       : Fintype P) ( $\_ : \text{DecidableEq } P$ )
1522       (e :  $\Theta \approx_o \text{LowerSet } P$ ),
1523       ( $\forall a, v a = \sum p \in (e a).$ toFinset, (v
1524         .mapEquiv e).mass p)  $\wedge$ 
1525        $\sum p, (v.$ mapEquiv e).mass p = 1
1526
1527   noncomputable def deltaPoint (p : P) :
1528     Valuation (LowerSet P)
1529
1530   theorem Valuation.eq_mix_deltaPoint (v :
1531     Valuation (LowerSet P)) :
1532     v = Valuation.mix v.mass v.sum_mass (
1533       fun p => deltaPoint p)
1534
1535   theorem
1536     exists_valuation_not_point_representable
1537     :
1538      $\exists v : \text{Valuation } (\text{LowerSet } \mathbb{N}), \sum' p, v.$ 
1539       mass p  $\neq v \top$ 
1540

```

```

1541
1542 theorem topIndicator_mass (n : ℕ) :
1543   topIndicator.mass n = 0
1544
1545 theorem topIndicator_not_scott :
1546   (⊔ n, topIndicator (LowerSet.Iic n)) ≠
1547     topIndicator T
1548
1549 theorem eq_tsum_mass_of_scott (v :
1550   Valuation (LowerSet ℕ))
1551   (hsc : v T = ⊔ n, v (LowerSet.Iic n))
1552     : v T = ∑' n, v.mass n
1553
1554 variable [V p : P, Finite (Set.Iic p)]
1555
1556 theorem Valuation.isPurelyAtomic_of_scott
1557   (v : Valuation (LowerSet P))
1558   (hsc : v T = ⊔ (S : Finset P), v (S.
1559     sup LowerSet.Iic)) : IsPurelyAtomic
1560     v
1561
1562 theorem tsum_mass_le (v : Valuation (
1563   LowerSet P)) : ∑' p, v.mass p ≤ v T
1564
1565
1566

```

B.4 The Dempster–Shafer Reading

```

1567 theorem Valuation.inclusion_exclusion (v :
1568   Valuation Ω) (s : Finset ι)
1569   (a : ι → Ω) : v (s.sup a) + v.iesEven
1570     s a = v.iesOdd s a
1571
1572 theorem Valuation.infty_monotone (v :
1573   Valuation Ω) (s : Finset ι) (a : ι → Ω)
1574   :
1575     v.iesOdd s a ≤ v ((s.sup a)cc) + v.
1576       iesEven s a
1577
1578 theorem Valuation.two_monotone (v :
1579   Valuation Ω) (a b : Ω) :
1580   v a + v b ≤ v (a ⊔ b)cc + v (a ⊓ b)
1581
1582 theorem Valuation.plausibility_sub_self (v
1583   : Valuation Ω) (a : Ω) :
1584   v.plausibility a - v a = v.slack a
1585
1586 theorem Valuation.sup_compl_decomp (v :
1587   Valuation Ω) (a b : Ω) :
1588   v (a ⊔ bc) = v bc + (v (a ⊓ b) + v.
1589     emGap a b)
1590
1591 noncomputable def Valuation.dempsterCond (
1592   v : Valuation Ω) (b : Ω)
1593   (hb : v b ≠ 0) : Valuation Ω
1594
1595

```

```

theorem Valuation.dempsterCond_apply (v :
  Valuation Ω) (b : Ω) (hb : v b ≠ 0)
  (a : Ω) : v.dempsterCond b hb a = (v (
    a ⊔ bc) - v bc) / (1 - v bc)
theorem Valuation.
  dempsterCond_eq_condVal_iff_slack (v :
  Valuation Ω) {b : Ω}
  (hb : v b ≠ 0) :
  (∀ a, v.dempsterCond b hb a = v.
    condVal b hb a) ↔ v.slack b = 0

```

B.5 Cox's Theorem

```

structure CoxModel (Ω : Type*) [Order.
  Frame Ω] where
  pl : Ω → Ω → ℝ
  mono_left : ∀ c, Monotone fun a => pl a
    c
  pl_bot : ∀ c, pl ⊥ c = 0
  pl_top : ∀ c, pl T c = 1
  F : ℝ → ℝ → ℝ
  conj : ∀ a b c, pl (a ⊓ b) c = F (pl a (
    b ⊓ c)) (pl b c)
  F_assoc : ∀ x y z, F (F x y) z = F x (F
    y z)
  F_strictMono_left : ∀ y, 0 < y →
    StrictMono fun x => F x y
theorem isDeMorgan_lowerSetV : Order.Frame
  .IsDeMorgan (LowerSet V)
theorem modularity_irreducible :
  ∃ (Ω : Type) (⊥ : Order.Frame Ω) (q :
    Ω → ℝ≥0∞),
    Monotone q ∧ q ⊥ = 0 ∧ q T = 1 ∧
    (∀ x y : Ω, x ⊓ y = ⊥ → q (x ⊔ y) =
      q x + q y) ∧
    ¬ (∀ x y : Ω, q x + q y = q (x ⊔ y)
      + q (x ⊓ y))
theorem no_disjunction_functional :
  ¬ ∃ S : ℝ≥0∞ → ℝ≥0∞ → ℝ≥0∞,
    ∀ x y : LowerSet V, qV (x ⊔ y) = S
      (qV x) (qV y)
theorem constructive_cox (M : CoxModel Ω)
  (hmod : ∀ a b : Ω, M.pl a T + M.pl b T
    = M.pl (a ⊔ b) T + M.pl (a ⊓ b) T)
  :
  ∃ (v : Valuation Ω) (g : ℝ → ℝ≥0∞),
    StrictMonoOn g (Set.Icc 0 1) ∧ g 0 =
      0 ∧ g 1 = 1 ∧
    ∀ a, v a = g (M.pl a T)

```

```

1651 noncomputable def coxModelENNReal :
1652   CoxModel  $\mathbb{R}_{\geq 0\infty}$ 
1653
1654 theorem Scale.exists_mul_generator :
1655    $\exists G : \mathbb{R} \rightarrow \mathbb{R},$ 
1656      $(\forall x\ y, S.u \leq x \rightarrow S.u \leq y \rightarrow G (S.F\ x$ 
1657        $y) = G\ x * G\ y) \wedge$ 
1658      $(\forall x\ y, S.u \leq x \rightarrow x < y \rightarrow G\ x < G\ y)$ 
1659
1660 theorem Scale.aczelStatement_cone :
1661    $\exists G : \mathbb{R} \rightarrow \mathbb{R},$ 
1662      $(\forall x\ y, S.u \leq x \rightarrow S.u \leq y \rightarrow G (S.F\ x$ 
1663        $y) = G\ x * G\ y) \wedge$ 
1664     StrictMonoOn G (Set.Ici S.u)
1665
1666 theorem exists_diag_eq {F :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ } {a
1667   b c :  $\mathbb{R}$ } (hab : a  $\leq$  b)
1668   (hcont : ContinuousOn (fun x => F x x)
1669     (Set.Icc a b))
1670   (hc : c  $\in$  Set.Icc (F a a) (F b b)) :  $\exists$ 
1671     x  $\in$  Set.Icc a b, F x x = c
1672
1673 structure Scale where
1674   F :  $\mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ 
1675   hdiv :  $\forall c, \exists x, F\ x\ x = c$ 
1676   hassoc :  $\forall x\ y\ z, F (F\ x\ y)\ z = F\ x (F\ y$ 
1677     z)
1678   hpos :  $\forall c\ x, x < F\ x\ c$ 
1679   hmono :  $\forall x_1\ x_2\ y_1\ y_2 : \mathbb{R}, x_1 \leq x_2 \rightarrow y_1$ 
1680      $\leq y_2 \rightarrow F\ x_1\ y_1 \leq F\ x_2\ y_2$ 
1681   hcont : Continuous fun p :  $\mathbb{R} \times \mathbb{R} \Rightarrow F\ p$ 
1682     .1 p.2
1683   hbot :  $\forall x, \text{Filter.Tendsto (fun z => F\ x$ 
1684     z) Filter.atBot (nhds x)
1685   u :  $\mathbb{R}$ 

```

B.6 Two Grounding Models

```

1689 noncomputable def MeasureTheory.Measure.
1690   toValuationOpens
1691   ( $\mu$  : Measure X) [IsProbabilityMeasure
1692      $\mu$ ] : Valuation (Opens X)
1693
1694 theorem toValuationOpens_slack_eq_frontier
1695   ( $\mu$  : Measure X)
1696   [IsProbabilityMeasure  $\mu$ ] (U : Opens X)
1697   :
1698      $\mu.toValuationOpens.slack\ U = \mu ($ 
1699       frontier (U : Set X))
1700
1701 theorem interiorMeasure_add_compl_le ( $\mu$  :
1702   Measure X) [IsProbabilityMeasure  $\mu$ ]
1703   (S : Set X) :  $\mu.interiorMeasure\ S + \mu.$ 
1704     interiorMeasure  $S^c \leq 1$ 

```

```

1706 theorem
1707   isDeMorgan_opens_iff_extremallyDisconnected
1708   :
1709     Order.Frame.IsDeMorgan (Opens X)  $\leftrightarrow$ 
1710       ExtremallyDisconnected X
1711
1712 theorem isDeMorgan_opens_iff_stonean [
1713   CompactSpace X] [T2Space X] :
1714     Order.Frame.IsDeMorgan (Opens X)  $\leftrightarrow$ 
1715       ExtremallyDisconnected X
1716
1717 inductive ThreePt : Type
1718   | ptA : ThreePt
1719   | ptB : ThreePt
1720   | ptZ : ThreePt
1721
1722 theorem frontier_subset_ptZ {U : Set
1723   ThreePt} (hU : IsOpen U) :
1724   frontier U  $\subseteq$  {ptZ}
1725
1726 theorem exists_ne_valuation_same_slack :
1727    $\exists v\ v' : \text{Valuation (Opens ThreePt)}, v.$ 
1728     slack = v'.slack  $\wedge v \neq v'$ 
1729
1730 theorem haltsOpen_compl : haltsOpenc = ( $\perp$ 
1731   : Opens Prop)
1732
1733 theorem slack_eq_boundary {p :  $\mathbb{R}_{\geq 0\infty}$ } (hp1
1734   : p  $\leq 1$ ) :
1735   (sierpinskiValuation hp1).slack
1736     haltsOpen
1737     = coinMeasure p (frontier {q : Prop
1738       | q})
1739
1740 theorem sharpReadout_not_computable (n :  $\mathbb{N}$ 
1741   ) :
1742      $\neg \text{ComputablePred fun c : Code =>}$ 
1743       sharpReadout n c haltsOpen = 1
1744
1745 theorem isOpen_wordCylinder (w : List Bool
1746   ) : IsOpen (wordCylinder w)
1747
1748 theorem cantorMeasure_wordCylinder (w :
1749   List Bool) :
1750   cantorMeasure (wordCylinder w) = (1 /
1751     2) ^ w.length
1752
1753 theorem
1754   disjoint_wordCylinder_of_not_prefix {v
1755   w : List Bool}
1756   (hv :  $\neg v <+ w$ ) (hw :  $\neg w <+ v$ ) :
1757   Disjoint (wordCylinder v) (
1758     wordCylinder w)

```

```

theorem cantorMeasure_biUnion_wordCylinder
  {F : Finset (List Bool)}
  (hF : PrefixFree F) :
  cantorMeasure (⋃ w ∈ F, wordCylinder w)
    = ∑ w ∈ F, (1 / 2 : ℝ≥0∞) ^ w.
    length

```

B.7 The Monad Question

```

def Valuation.comap (f : FrameHom Γ Ω) (v
  : Valuation Ω) : Valuation Γ

theorem Valuation.comap_id (v : Valuation
  Ω) : v.comap (FrameHom.id Ω) = v

theorem Valuation.comap_comp (f : FrameHom
  Γ Ω) (g : FrameHom Δ Γ)
  (v : Valuation Ω) : v.comap (f.comp g)
    = (v.comap f).comap g

theorem exists_rect_decomposition (W :
  LowerSet (ℕ × ℕ)) :
  ∃ (N : ℕ) (A B : ℕ → LowerSet ℕ),
    W = (Finset.range N).sup fun i => A
      i ×s B i

theorem Valuation.eq_on_finsetSup_of_eq_on
  {G : Set Ω}
  (hG : ∀ g ∈ G, ∀ g' ∈ G, g ⊔ g' ∈ G) (
  u u' : Valuation Ω)
  (h : ∀ g ∈ G, u g = u' g) {ι : Type*}
    [DecidableEq ι]
  (s : Finset ι) (a : ι → Ω) (ha : ∀ i ∈
    s, a i ∈ G) :
  u (s.sup a) = u' (s.sup a)

theorem Valuation.product_unique (v w :
  Valuation (LowerSet ℕ))
  {u u' : Valuation (LowerSet (ℕ × ℕ))}
  (hu : ∀ A B : LowerSet ℕ, u (A ×s B) =
    v A * w B)
  (hu' : ∀ A B : LowerSet ℕ, u' (A ×s B)
    = v A * w B) : u = u'

noncomputable def ultrafilterValuation (W
  : Ultrafilter ι) : Valuation (Set ι)

theorem rect_mem_iff {α β : Type*} {W :
  Ultrafilter (α × β)}
  {S : Set α} {T : Set β} :
  S ×s T ∈ W ↔ S ∈ W.map Prod.fst ∧ T ∈
    W.map Prod.snd

theorem product_valuation_not_unique :

```

```

  ∃ u u' : Valuation (Set (ℕ × ℕ)),
    (∀ S T : Set ℕ, u (S ×s T) = u' (S ×
      s T)) ∧ u ≠ u'

```

```

theorem fubini_fails_for_valuations :
  ∃ (v w : Valuation (Set ℕ)) (u u' :
  Valuation (Set (ℕ × ℕ))),
    (∀ S T : Set ℕ, u (S ×s T) = v S * w
      T) ∧
    (∀ S T : Set ℕ, u' (S ×s T) = v S *
      w T) ∧ u ≠ u'

```

```

noncomputable def Valuation.mixCountable {
  ι : Type*} (w : ι → ℝ≥0∞)
  (hw : ∑' i, w i = 1) (v : ι →
  Valuation Ω) : Valuation Ω

```

```

theorem Valuation.
  mixCountable_mixCountable {ι : Type*} {
  κ : ι → Type*}
  (w : ι → ℝ≥0∞) (hw : ∑' i, w i = 1)
  (u : ∀ i, κ i → ℝ≥0∞) (hu : ∀ i, ∑' j,
    u i j = 1)
  (v : ∀ i, κ i → Valuation Ω) :
  Valuation.mixCountable w hw
    (fun i => Valuation.mixCountable (
      u i) (hu i) (v i))
    = Valuation.mixCountable (fun p : Σ
      i, κ i => w p.1 * u p.1 p.2)
      (tsum_sigma_mul_eq_one w hw u hu)
      (fun p => v p.1 p.2)

```

```

theorem Valuation.mixCountable_unique (v :
  Valuation Ω) :
  Valuation.mixCountable (fun _ : PUnit
    => 1) (by simp) (fun _ => v) = v

```

```

theorem Valuation.mixCountable_comm {ι κ :
  Type*}
  (w : ι → ℝ≥0∞) (hw : ∑' i, w i = 1) (u
    : κ → ℝ≥0∞) (hu : ∑' j, u j = 1)
  (v : ι → κ → Valuation Ω) :
  Valuation.mixCountable w hw (fun i =>
    Valuation.mixCountable u hu (v i))
    = Valuation.mixCountable u hu
      (fun j => Valuation.mixCountable
        w hw (fun i => v i j))

```

C The Aczél Development

This appendix sketches the from-scratch construction behind the product-rule half of Theorem 30. The target shape is Aczél's associativity theorem [Aczél 1966]: a suitably regular associative $F : \mathbb{R} \rightarrow \mathbb{R} \rightarrow \mathbb{R}$ reg graduates to multiplication,

$G(Fx y) = G(x) \cdot G(y)$ for all x, y and a strictly monotone G . `mathlib` has no associativity representation, no Hölder-style embedding of ordered semigroups on intervals, and no Cauchy functional equation under monotonicity (only under continuity).

Hypotheses. A Scale bundles: associativity of F , *divisibility* $\forall c \exists x, Fx x = c$, positivity on the cone $x < Fx c$, joint order-preservation, joint continuity, a unit u , and identity at $-\infty$, meaning $Fx z \rightarrow x$ as $z \rightarrow -\infty$. Order-preservation was absent from our first bundle, and the additivity proof needs it. Divisibility supplies the 2^n -th roots that the dyadic construction below counts, and without it the generator cannot be written down. It is taken as a plain hypothesis in the bundle, although its interval form is a consequence of the intermediate value theorem (`exists_diag_eq`). Non-vacuity: *log-sum-exp*, $Fx y = \log(e^x + e^y)$ with $u = 0$, satisfies every field (`logSumExpScale`), and for it the intended generator is globally `exp`.

Construction. Write $a^{(n)}$ for n right-associated copies of a under F (`Fpow`). Associativity alone makes $n \mapsto a^{(n)}$ a semigroup morphism from $(\mathbb{N}, +)$, with no commutativity needed. Commutativity is a *corollary* of the generator, not a prerequisite. One fact usually assumed is here *derived*: unboundedness of the iterates. Were they bounded, monotone convergence would give a fixed point $FLa = L$, contradicting positivity, and unboundedness yields the Archimedean property. Divisibility supplies a tower of 2^n -th roots $u \succ u^{1/2} \succ u^{1/4} \succ \dots$ (`droot`) with the coherence law that 2^ℓ fine copies recombine one coarse mark. Counting how many copies of the n -th root fit under b (`dcount`, well defined by the Archimedean property) gives dyadic approximants $g_n(b) = \text{dcount}(b, n)/2^n$. The refinement sandwich $2N + 1 \leq N' \leq 2N + 2$, where N and N' are the counts at consecutive refinement levels n and $n + 1$, makes them monotone and Cauchy, and the generator is their limit, $g(b) = \bigsqcup_n g_n(b)$.

Results (all sorry-free, standard axioms). On the cone $\{x \mid u \leq x\}$ we prove three things. Normalization: $g(u) = 1$. Additivity: $g(Fx y) = g(x) + g(y)$, by squeezing super- and sub-additive counting bounds (`dcount` $x + \text{dcount } y + 1 \leq \text{dcount}(Fx y) \leq \text{dcount } x + \text{dcount } y + 3$) against a vanishing 2^{-n} error term. And *strict* monotonicity, proved without any continuity of g by a count-amplification argument: a shrinking root inserted between $x < y$ contributes a full extra block of 2^{n-M} copies, M being the refinement level of the inserted root. Exponentiating gives the multiplicative form, a strictly monotone G with $G(Fx y) = G(x)G(y)$ on the cone (`exists_mul_generator`), which is the conclusion shape of Aczél's theorem restricted to the cone (`aczelStatement_cone`). A bounded reorientation $\bar{G} = \exp(-g)$ transports it to the $[0, 1]$ -valued conjunctive orientation of Section 6.

Remaining gap. The constructed g is discontinuous at the unit (it is 0 below u), so matching the classical statement verbatim still needs the off-cone extension (a group completion) and continuity of the extension. That is genuine analysis, open here, but not vacuous: for the log-sum-exp example the global continuous generator exists (`hasOrderedGenerator_logSumExp`). Of Hölder's classical hypotheses, the Archimedean property, which rules out elements too fine for any real number to detect and is what makes an embedding into $\mathbb{R}$ possible at all, is a theorem in this development rather than an assumption, and the interval form of divisibility follows from the intermediate value theorem. Only the embedding's global regularity remains.